

Course Programme

ETS052 Computer Communication

Introduction

Welcome to the course Computer communication. The objective of this course is to give you an introduction to the area of data communication and networking. This course includes everything from how to send bits on a link, to how Internet works. The course consists of 10 lectures, 6 problem solving classes, 2 laboratory sessions, 2 midterm exams, and a final exam. All information about this course can also be found on the course's webpage:

<http://www.telecom.lth.se/Kurser/datorkomED>.

Lectures

The course begins on Monday October 23 at 13.15 in Aulan (Kårhuset). The course consists of 10 lectures, according to the preliminary schedule shown below. All lectures start at 13.15. The first lecture will be given in English. The rest of the lectures will be given in Swedish, with English slides.

No.	Date	Place	Content
1	23/10	Aulan	Introduction, Data and signals, Transmission media
2	26/10	E:A	Digital and Analog transmission, Multiplexing and Spreading
3	30/10	Aulan	Error detection and correction, Data link control
4	2/11	E:A	HDLC and PPP, Multiple access
5	6/11	Aulan	Ethernet, WLAN and Bluetooth, VLAN
6	9/11	E:A	Switching and WAN, Layered network models
7	16/11	E:A	IP, Address mapping and error reporting, DNS, UDP and TCP
8	20/11	Aulan	Internet routing
9	23/11	E:A	Internet Security, Performance issues
10	27/11	Aulan	SUNET and LUNET

Problem solving classes

The course includes six problem solving classes. The classes start Tuesday October 31. The classes are scheduled as follows:

Time	Room	Extra room
Tuesdays 8-10	E:3315	
Tuesdays 10-12	E:3318	E:3316
Tuesdays 13-15	E:0522	E:1147
Tuesdays 15-17	E:3316 (in English)	E:3315

After the first week of classes, we will decide which of the extra rooms that will be used.

Laboratory sessions

The course includes two mandatory laboratory sessions. You need to sign up for each session! The session lists will be posted on the course noteboard outside the department. The schedule is as follows:

Lab session	Scheduled sessions	Lists posted
PPP	13/11-24/11	30/10 at 12.30
Networking	27/11-8/12	13/11 at 12.30

Exams

There are two voluntary midterm exams in the course. Each midterm exam is worth 10 credits, which you then can use in the final exam. The dates and contents for the midterm exams are shown below::

Midterm exam	Date and place	Content
1	13/11 13.15: Aulan	Lectures 1-5, Problem solving classes 1 and 2
2	4/12 13.15: Aulan	Lectures 6-10, Problem solving classes 3-5

The final exam is held **14/12 at 14.00-19.00 in MA:10**. The final exam contains 100 credits and you need 50 credits to pass the exam.

Course Literature

There are two alternative textbooks in the course:

1. "Data Communication and Networking", B A Forouzan, McGraw Hill, 4th edition.
2. "Datakommunikation - en inledande översikt", M Kihl, Studentlitteratur, 2nd edition. This book is used together with the report "Routern och routing - en introduktion" by Jens A Andersson, which is sold at the department.

Also, the material for the problem solving classes and laboratory session 2 (Networking) is sold at the department.

Involved Personnel

Maria Kihl, 046-222 9010, maria@telecom.lth.se (main teacher)
Jens A Andersson, jason@telecom.lth.se (assistant lecturer)
Jens K Andersson, jensa@telecom.lth.se (teaching assistant)
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Course secretary Marianne Greiff Svensson, Room 5136:D, is available Monday-Friday 10-12.